

Aaryan Singh

Software Engineer | Product, AI Agents & Systems

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Waterloo dropout. Hacked rice.edu at 17 (responsibly disclosed SQL injection) and Waterloo's GPU cluster. Early engineer at Extraordinary; founding engineer at ReferRush. Built a Rust service benchmarked at **1.34M req/s** (four-vCPU loopback, pipeline 512).

EXPERIENCE

Founding Engineer | ReferRush | Feb 2025 - Sep 2026 | Bengaluru

- Built Shopify installation, WhatsApp referrals and UPI payouts from the first commit, including PAN and TDS handling. The platform reached 2,300 merchants, 30M+ customers and \$1M+ processed.
- Diagnosed a bank-reversal loop through payment-ID logs. Added a two-minute watchdog, EC2 worker recovery, alerts and payment isolation so other payouts could continue.
- Worked directly with customers and closed brands through sales.

Backend Engineer | O1Visa | Jan 2024 - Jan 2025

- Built indexes, search and the customer directory powering onboarding for an O-1 visa platform.

Founder & Software Engineer | FnBC.ltd | Oct 2022 - Nov 2024 | Remote (Dubai)

- Ran a freelance software agency alongside employment, building AI agents and full-stack products for clients.

Software Engineer | MailTime Email Client | Oct 2022 - Sep 2024

- Engineered email indexing, search and automation.

SELECTED PROJECTS

instantKV | Rust, Tokio, Axum, redb, MCP. Built local-first agent memory in Rust: atomic topic/tag/time indexes, bounded BM25 search, revisioned writes and native OpenCode MCP. Measured 95.13% Recall@10 across all 500 LongMemEval-S questions with 14.34 MiB sampled server RAM on an M4 Pro. QA: 85.20% with GPT-6 Luna reader/judge (model variant). Source MVP; phone integration is planned.

Morse | Rust agent runtime. Built remote execution that continues after client disconnects, streamed tool output, persisted sessions, event replay and crash recovery. A separate side agent reports progress without interrupting active work.

FireScope | TypeScript Firebase SDK & CLI. Built typed Functions/Firestore helpers, generated configuration and staged rebuilds. Verified HTTP, Hosting, callable functions and Firestore triggers in real emulators; added Node.js 22/24 CI.

TiVM | Python computer-use testing. Built Linux desktop testing through accessibility, screenshots, video and reports. Added authenticated APIs, commit-pinned jobs, atomic queue claims, webhook deduplication and restart recovery.

rushort | Rust HTTP/1.1, Tokio, SQLite WAL. Built a raw HTTP service with commit-before-ack writes, live backups and rollback. Measured 1.34M redirects/s on a four-vCPU VPS: loopback, pipeline depth 512, median of three eight-second runs, zero errors.

Rove | Rust terminal AI assistant. Built streamed model/tool output, a live observer feed, bearer authentication and process-group timeouts; deployed verified commits through CI with rollback.

OPEN SOURCE

Contributor: 47 merged PRs across 13 repositories. Atuin (31,876 stars), Podman Desktop (8,046 stars), WebTUI (2,447 stars), Nimbus (1,967 stars), iamb (1,298 stars); 27 PRs in termLens. Star counts: Oct 2, 2026.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Python, Go, Swift, Shell

Product & AI: React, Next.js, Firebase, AI agents, MCP, computer-use testing

Systems & infrastructure: Tokio, Axum, redb, HTTP, WebSocket, SSE, SQLite, PostgreSQL, AWS EC2, Docker, Linux, GitHub Actions, Cloudflare Workers, R2, Queues

EDUCATION & RECOGNITION

University of Waterloo dropout | Computer Engineering studies | Left to build full time

3x National Math Olympiad silver (ages 15–17); IITM NexHack 2nd (2023); IPSC Web Dev 4th in India.